

Hamro Hospital - Real Hospital Deployment Guide

Prepared for final production delivery. This document is suitable for technical teams, hospital management, administrators, and deployment engineers.

1. Hospital Deployment Overview

This guide explains how a real hospital can deploy Hamro Hospital Management System. It is written for hospital administrators, IT teams, and implementation partners.

2. Required Hardware

Scale	Recommended Server
Small hospital	8 CPU cores, 16-32 GB RAM, 1 TB SSD
Medium hospital	16 CPU cores, 64 GB RAM, 2 TB NVMe SSD
Large hospital	Multiple app servers, dedicated PostgreSQL server, 128 GB+ RAM
Enterprise	Load balancer, app cluster, DB replication, separate backup server

3. LAN and Network Setup

- Use gigabit LAN minimum for counters and departments.
- Use managed switches for large buildings.
- Keep server on static LAN IP.
- Separate guest Wi-Fi from hospital system network.
- Use UPS for server, network switch, receipt printers, and barcode counters.

4. PostgreSQL Production Deployment

- Install PostgreSQL 15/16.
- Create database and user.
- Set DJANGO_DB_ENGINE=postgres.
- Run migrations.
- Enable backups and monitoring.
- Tune shared_buffers, work_mem, maintenance_work_mem, and max_connections according to RAM.

5. Printer and Barcode Setup

- Use USB/LAN thermal receipt printer for cash counter.
- Use A4 laser printer for reports, lab/radiology, admissions, and official records.
- Use standard keyboard-emulation barcode scanners.
- Ensure barcode scanner sends Enter after scan where forms require auto-submit.

6. Initial Hospital Setup

- Configure hospital name, logo, address, PAN/registration, and receipt settings.
- Create departments, units, doctors, counters, department heads, finance users, and staff.
- Configure extension service doctors and fees.
- Configure wards, beds, services, radiology/lab/pharmacy master data.

7. Daily Operation

- Registration creates/looks up patients by barcode/phone/name.
- Doctor consults and sends referrals.
- Departments process orders.
- Cash counter collects payments.
- Finance reviews revenue and payroll.
- Super Admin monitors reports, audit, backup, and staff workflows.

8. Backup and Recovery Procedure

- Run database backup daily.
- Run full database+media backup at least daily.
- Sync backup folder to secure cloud/NAS.
- Test restore in staging monthly.
- Never restore directly on live system without verified downtime and approval.

9. Software Update Procedure

- Back up database and media.
- Pull latest code.
- Install updated requirements.
- Run migrations.
- Run checks/tests.
- Collect static files.
- Restart application services.
- Verify dashboards and reports.

10. Scalability Matrix

Concurrent Users	CPU	RAM	Storage	Notes
500	8-12 cores	32 GB	1-2 TB NVMe	Single app + DB possible
1,000	16 cores	64 GB	2 TB NVMe	Separate DB recommended
5,000	3+ app servers + DB	128-256 GB DB	4-8 TB NVMe	Load balancer, Redis, read replicas
10,000	App cluster + HA DB	256 GB+ DB	10 TB+ enterprise SSD	Dedicated DevOps, monitoring, HA/DR

11. Approximate Nepal Deployment Costs

These are approximate estimates only and must be validated before procurement.

Item	Approximate Cost NPR
Local server small/medium	250,000 - 900,000
Enterprise local server	1,200,000 - 5,000,000+
Cloud hosting small	8,000 - 30,000/month
Cloud hosting medium/large	40,000 - 300,000+/month
Backup storage	1,000 - 25,000/month
Domain	1,500 - 5,000/year
SSL	Free Let's Encrypt or 5,000 - 50,000/year
SMS/OTP	0.8 - 3 NPR per SMS depending provider
Maintenance	30,000 - 300,000+/month depending SLA

12. Security Best Practices

- Use HTTPS.
- Use strong admin passwords.
- Restrict server SSH access.
- Keep .env private.
- Schedule backups.
- Review audit logs.
- Use least-privilege staff roles.
- Patch OS and dependencies regularly.

13. Registration Counter Setup Checklist

- Create user accounts for Registration Counter.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

14. Cash Counter Setup Checklist

- Create user accounts for Cash Counter.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

15. Doctor Rooms Setup Checklist

- Create user accounts for Doctor Rooms.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

16. Laboratory Setup Checklist

- Create user accounts for Laboratory.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

17. Radiology Setup Checklist

- Create user accounts for Radiology.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

18. Pharmacy Setup Checklist

- Create user accounts for Pharmacy.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

19. Ward/Admission Setup Checklist

- Create user accounts for Ward/Admission.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

20. Nursing Station Setup Checklist

- Create user accounts for Nursing Station.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

21. Operation Theatre Setup Checklist

- Create user accounts for Operation Theatre.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

22. Blood Bank Setup Checklist

- Create user accounts for Blood Bank.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

23. Insurance Counter Setup Checklist

- Create user accounts for Insurance Counter.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

24. Finance Office Setup Checklist

- Create user accounts for Finance Office.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

25. Super Admin Office Setup Checklist

- Create user accounts for Super Admin Office.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.

26. IT/Server Room Setup Checklist

- Create user accounts for IT/Server Room.
- Verify printer/barcode/network access where applicable.
- Test login and dashboard access.
- Perform sample workflow in staging before go-live.
- Train users on daily process and escalation path.